Analogue Electronics

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Introduction

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1. Course Information

1. Course Code: M201105W

2. Course Category: Specialty Basic

3. Course Nature： Compulsory

4. Hours / Credits: 48/3

5. Prerequisite(s):

College Physics, Circuits Analysis, Advanced AC Circuits

6. Follow-up courses :

Advanced communication systems, Individual BEng project

7. Specialty: Communication Engineering

2. Course Description and Objectives

2.1 Description

Through the study of this course, students can

- **deeply understand relevant theories.**
- **grasp and improve the ability to analyze and evaluate analogue circuits and systems qualitatively and quantitatively.**
- **cultivate engineering practice ability of identifying, solving and assessing questions, which are required by industry both in the context of analogue circuits and in the wider engineering discipline, and lay a good foundation for further study of follow-up courses.**

2. Course Description and Objectives

2.2 Objectives

On successful completion of this course students will be able to:

① be aware of the wide extent of analogue components, and understand basic concepts and theories of semiconductor devices;

② master circuit analysis methods in time and frequency domains;

③ master analysis and evaluating methods for fundamental analogue circuit building blocks;

2. Course Description and Objectives

2.2 Objectives

On successful completion of this course students will be able to:

④ **master basic skills of resourcing IC manuals and related information;**

⑤ **master the basic experimental design and test skills of analogue circuits;**

⑥ **improve engineering practice ability of finding, solving and assessing problems.**

3. Role of the Course

Science

Ideas & Needs

Analog & Digital Electronic Technology

Engineering

Products

3. Role of the Course





- System-Board-Circuits-Component-Material
- Function-Performance-Power Dissipation

3. Role of the Course

For Product Development

- **Identifying the Need**
- **Designing**
- **Prototyping and Evaluating**
- **Producing**
- **Marketing**
- **Servicing**

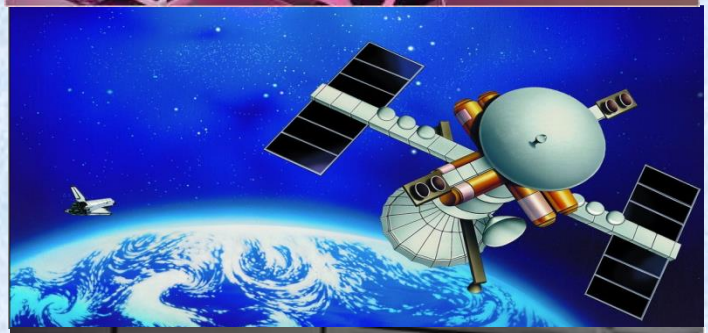


3. Role of the Course

For Applications areas

- Communication System
- Control System
- Testing System
- Computers
- Family Electronics
- Agriculture Machine
- Biomedical Engineering
- Spaceflight and Airplane
- …







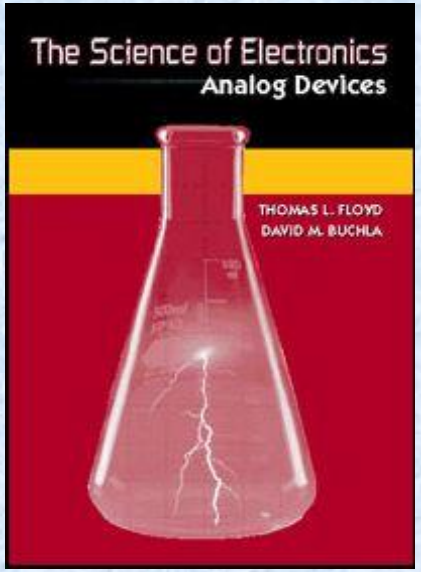
4. Course Overview

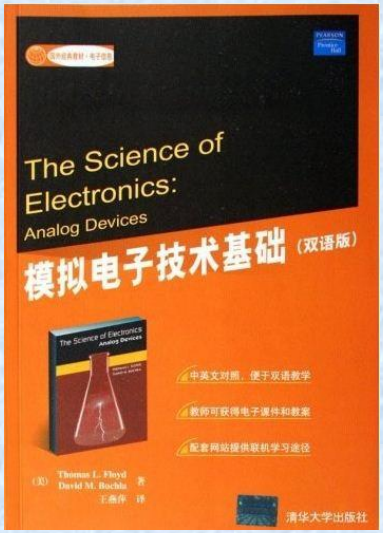
- Section 1 Overview of Electronic Technology
- Section 2 Semiconductor Devices
- Section 3 Single-stage Amplifiers
- Section 4 Multi-stage Amplifiers
- Section 5 Differential Amplifiers
- Section 6 Power Amplifiers
- Section 7 Frequency Response of Amplifier Circuits
- Section 8 Feedback in Amplifier Circuits
- Section 9 Operational Amplifiers and Applications
- Section 10 Oscillators
- Section 11 DC Power Supply

5. Textbook

Original Version

- The Science of Electronics: Analog Devices
- Thomas L.Floyd， David M. Buchla.
- Pearson Education, Inc.





Double Language Version

- 模拟电子技术基础（双语版）
- Thomas L.Floyd，David M. Buchla.
- 清华大学出版社

6. Recommended Books

[1] Mahmood Nahvi, Joseph A. Edminister. Schaum's Outline of Theory and Problems of Electric Circuits, 4th ed., New York, USA: McGraw-Hill, 2003.
[2] Donald A. Neamen. Electronic Circuit Analysis and Design, 4nd ed., Boston, Mass. , USA: McGraw-Hill, 2018.
[3] Ying Liu. Analog Electronic Technology. Beijing: Tsinghua University Press & Beijing Jiaotong University Press, 2008.
[4] Yong Lu. Foundation of analog integrated circuits. 3rd ed.. Beijing: China Railway Publishing House, 2010.
[5] Arthur B. Williams, Fred J. Taylor. Electronic Filter Design Handbook, 4th ed., New York, USA: McGraw-Hill, 2006.
[6] Douglas Self. Audio Power Amplifier Design, 6th ed., New York, UK: Focal Press, Taylor & Francis Group, 2013.

7. Reference Websites

- Course Learning
  http://www.prenhall.com/SOE

- Component Data Sheets Downloading
  http://www.21icsearch.com
  http://www.ic-on-line.cn
  http://www.icpdf.com
  http://www.ic-on-line.net

- Comprehensive Websites of Electronic Technology
  http://www.21ic.com
  http://www.ednchina.com
  http://www.eefocus.com
  http://www.ourdev.cn

8. Schedule

- **Class-teaching**
- **Coursework**
- **Experiments**
- **Tutorials**

Course Calendar

9. Assessments

Final Score = **coursework** (20%) + **Final-exam** (80%)

Coursework = Labs (10%) + homework (10%)

lab report lab work

assignments assignments

10. Others

Coursework Guideline

- Be assigned on the experiments and after the lectures related to the subject. See in Moodle
- Be due by the deadline: See in Moodle
- Submission & marking: Done by Moodle

Lecture Guideline

- Lecture notes will be available on Moodle

10. Others

Methods

- Lecture-based with emphasis on problem solving
- My responsibility:
  - ✓ Provide a cohesive set of lectures
  - ✓ Provide with motivation
  - ✓ Give you plenty of practice problems
- Your responsibility:
  - ✓ Turn up on lectures!
  - ✓ Complete all coursework
  - ✓ If unsure… Ask!